

BUSINESS PROPOSAL

UMHackathon 2026

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1. Executive Summary:

SCADA-i is a multi-agent control system built to transform university campuses. It delivers autonomous smart building management through a unified web dashboard and mobile app ecosystem. The platform connects estate managers, administrators, and students directly to facility operations. This empowers intelligent decision-making for lecture hall climate control, tutorial room scheduling, and sustainability tracking. Currently, Malaysian public (IPTA) and private (IPTS) universities face tight MOHE budgets and strict ESG compliance standards where this pressure creates a massive demand for cost-effective and energy-efficient campus operations. SCADA-i solves this with a software-first AI architecture by iterating into existing university infrastructure to eliminate the need for capital-intensive hardware overhauls. This intelligent approach drives predictive energy optimization (e.g. autonomously pre-cools lecture halls before classes begin, automatically turning off air conditioning units in empty rooms). By consolidating fragmented facility systems into a single intelligent ecosystem, SCADA-i is able to provide a modern campus experience that delivers immediate electricity cost reductions and massive operational efficiency at scale.

2. Product Overview / Solution

Key Features:

- *Role-specific interfaces featuring a web dashboard for admins/technicians to monitor operations, alongside a mobile application for room occupants to easily book spaces and adjust appliance settings.*
- *Real-time monitoring and analytics from live IoT data streams, including active occupancy feeds and room bookings.*
- *Smart orchestrator to handle user request by leveraging multiple working agents to the direction of optimizing energy and compliance to Environment Social Governance (ESG)*
- *Autonomous energy optimization through a specialized HVAC AI agent that handles pre-cooling, scheduling, and user-requested temperature overrides*
- *Intelligent ESG compliance tracking featuring space utilization analysis, carbon offset cost calculations, and PDF policy ingestion.*
- *Seamless facility scheduling driven by a Booking agent that manages room availability, resources, and user schedules.*

Technology Used:

- *LangGraph framework powering the multi-agent AI architecture (Orchestrator, HVAC, Booking, and ESG agents) for complex decision-making*
- *MQTT protocol facilitating reliable, real-time communication, command execution, and data replay with simulated IoT devices and sensors*
- *PostgreSQL and Supabase integration serving as the backend infrastructure for robust state management and agent history tracking*

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- *React and Vite driving the high-performance frontend interfaces, supported by automated CI/CD pipelines and guardrail safety checks via GitHub Actions*

3. Market Research & Market Dynamics

3.1 Problem Landscape

Modern facility managers and commercial building owners are overwhelmed by the sheer volume of IoT data generated by smart buildings, yet lack the automated tools to translate this raw data into actionable insights. This bottleneck is exacerbated by legacy SCADA systems that only provide raw metrics, forcing human operators into tedious manual analysis to catch inefficiencies. In the long run, facilities suffer from massive energy waste due to blind, fixed-schedule operations and will eventually face costly unexpected equipment breakdowns due to reactive rather than predictive maintenance. From an administrative standpoint, this reliance on manual intervention creates heavy operational workloads, increases the risk of human error, and fosters poor, fragmented coordination between distinct management departments such as maintenance, sustainability, and tenant administration.

3.2 Market Niche

SCADA-i targets commercial real estate developers, enterprise facility management companies, and corporate office operators, particularly within developing urban centers and smart cities. This core segment consists of stakeholders who urgently prioritize energy efficiency, significant operational cost reduction, and strict, automated compliance with emerging ESG (Environmental, Social, and Governance) reporting standards. On the operational level, the target niche focuses on building administrators and technicians seeking to automate their administrative workloads and transition toward AI-driven, predictive facility management. Furthermore, the platform serves the modern corporate tenant and employee who demands a seamless, digitally integrated workspace experience, prioritizing convenience in room scheduling and localized climate comfort.

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)
TAM	Global smart campus solutions market	RM 616 billion
SAM	Asia-Pacific / ASEAN smart campus infrastructure market	RM 158 - 202 billion
SOM	Realistic market capture (first 1.5-3 years, Malaysia + early SEA expansion targeting smart campuses)	RM 60 - 120 million

3.4 Market Dynamics & Placement:

There were few key dynamic that shape SCADA-i entry to this market:

- **IoT & Smart Sensor Proliferation:** *The rapid deployment and decreasing cost of IoT devices in commercial real estate have created a heavily connected, data-rich environment. This widespread sensor adoption provides the foundational infrastructure required for SCADA-i to continuously monitor and optimize building operations at a granular level.*
- **Rising ESG & Regulatory Mandates:** *Stricter environmental regulations and corporate sustainability goals are forcing building owners to actively monitor and reduce their carbon footprint. This regulatory pressure creates immediate, mandatory market demand for intelligent systems that can automatically track space utilization, optimize HVAC energy consumption, and generate actionable ESG compliance reports.*
- **Shift to AI-Driven Facility Management:** *Building administrators are increasingly burdened by the limitations of traditional, static SCADA systems and reactive maintenance schedules. There is a strong industry push to modernize operations by moving away from manual monitoring toward proactive, AI-driven automation and natural language interfaces that reduce administrative workloads and human error.*

4. Value Proposition & Competitive Advantage

SCADA-i is redefining the economics of smart building management by layering an intelligent, multi-agent AI framework (LangGraph) over a facility's existing IoT sensor infrastructure. This eliminates the need for costly "rip-and-replace" hardware upgrades. Compared to traditional SCADA systems, SCADA-i gains a massive competitive advantage by actively lowering energy expenditures through predictive HVAC optimization and drastically reducing the administrative overhead associated with manual data analysis and ESG reporting.

This software-first, AI-driven approach negates the need for continuous human monitoring, turning raw data into autonomous action. Furthermore, seamless integration of a technician-focused Web Dashboard and a tenant-focused Mobile App removes the friction traditionally associated with fragmented facility management tools. This end-to-end ecosystem—capable of executing complex, natural-language commands via specialized AI agents—creates a technological moat that legacy dashboard competitors cannot replicate without significant investments in generative AI and system orchestration.

5. Competitor Analysis (Porter's Five Forces)

Force	Intensity	Assessment
Competitive Rivalry	High	The Malaysian energy management and facility scheduling market is highly contested. Companies like Tanand Technology dominate AI-driven building automation, while ICEE focuses on heavy chiller-plant optimizations. Meanwhile, systems like eFACiLiTY capture the booking software market. SCADA-i differentiates by sitting right in the middle: using a multi-agent AI architecture (LangGraph) to actively bridge the gap between academic scheduling and autonomous HVAC control.
Threat of New Entrants	Medium	Because SCADA-i operates on a "software-first" overlay model using standard protocols (MQTT, APIs), the barrier to entry is lower than hardware manufacturing. Other software startups can enter this space easily. SCADA-i must build a defensive moat quickly through proprietary data collection, early enterprise partnerships in the SEA

		market, and superior natural-language orchestration that competitors take years to build.
Bargaining Power of Buyers	High	Malaysian public universities (IPTA) operate on tight Ministry of Higher Education (MOHE) budgets, while private universities (IPTs) demand immediate ROI. Buyers have significant leverage. SCADA-i counters this by framing the system not as an IT expense, but as a direct cost-saving tool—proving ROI through immediate reductions in Tenaga Nasional Berhad (TNB) Maximum Demand (MD) charges and aiding compliance with the newly passed Energy Efficiency and Conservation Act (EECA).
Bargaining Power of Suppliers	Medium	SCADA-i relies on cloud infrastructure (Supabase/PostgreSQL), LLM providers (e.g., OpenAI, Google Gemini), and off-the-shelf IoT sensors. Because IoT sensors are highly commoditized in the Southeast Asian supply chain and there is fierce competition among cloud/AI providers, suppliers have very little leverage over SCADA-i's pricing or operations.
Threat of Substitutes	Medium	The primary substitute is the "status quo": relying on university clerks to manage bookings via legacy portals or Google Sheets, while technicians and security guards manually turn off AC units in empty lecture halls. While this manual substitute avoids software fees, it results in massive energy waste and fails to meet modern university ESG ranking standards.

Competitor Comparison Table:

<i>Competitor</i>	<i>Strength</i>	<i>Weakness</i>	<i>SCADA-i Advantage</i>
eFACiLiTY Malaysia	World-class facility and meeting room booking platform. Excellent user interface, Outlook integration, and kiosk displays.	Primary focus is on scheduling. While they can trigger lights/AC to turn off after a booking, they rely on rigid "If/Then" rules rather than dynamic, predictive climate control.	Predictive Pre-Cooling: Leveraging LangGraph agents, SCADA-i proactively optimizes indoor comfort by analyzing class schedules and real-time weather data. Instead of reacting to occupancy, the system autonomously initiates pre-cooling approximately 15 minutes in advance, ensuring an optimal environment upon arrival while improving energy efficiency.
ICEE Energy Services	Deep engineering expertise. Their "iCEE Dynamics" IoT platform heavily optimizes massive central air-conditioning chiller plants for maximum energy savings	Highly CapEx-intensive or requires complex Energy Performance Contracts (EPC). Focuses purely on back-end chillers, completely invisible to room occupants.	Non-Intrusive Software Integration: SCADA-i functions as a room-level overlay (AHUs/FCUs/Split units) that integrates with existing edge sensors via MQTT, bypassing the need for capital-intensive hardware replacements or complex engineering overhauls.

6. Business Model Canvas

6.1. Customer Segmentation

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1. **Campus Estate Managers & Technical Operators** *University facility divisions and administrative clerks who require an intelligent, centralized web dashboard to automate tedious campus monitoring. SCADA-i streamlines daily operations—eliminating the need to physically patrol empty lecture halls to turn off ACs—and proactively addresses HVAC inefficiencies without technicians needing to constantly stare at complex, raw SCADA screens.*
2. **University Administration & Financial Leadership:** *Vice-Chancellors, Bursars, and executive leadership who must prioritize tight Ministry of Higher Education (MOHE) budgets and long-term campus ROI. They require automated tracking of faculty energy usage, immediate reduction of TNB electricity bills through predictive HVAC optimization, and the reliable generation of data for mandatory ESG compliance and global sustainability rankings.*
3. **Students, Lecturers & Campus Societies:** *The core campus population who value a seamless, mobile-friendly experience to instantly book tutorial rooms, and check the availability of shared resource centers, ensuring optimal room temperatures are set before their bookings even begin.*

6.2. Revenue Stream

Primary: B2B SaaS Subscription & Managed Hosting (Tiered based on campus size and connected spatial zones – Approx. RM 2,500 - RM 25,000 / month per faculty or campus). This includes a recurring customization and dedicated hosting fee for providing the university with a white-labeled, tailored mobile app and centralized web dashboard.

Secondary:

- **Onboarding & Integration Fee:** Complexity-based RM 15,000 - RM 50,000 / deployment for mapping existing IoT sensors and integrating with legacy academic timetables.
- **Hardware Partner Commission: Approx.** 10% - 15% commission of the order value when universities purchase necessary smart sensor upgrades through our referred local IoT vendors.

6.3. Cost Structure

Cost Drivers	Cost Type
● SCADA-i's maintenance of the LangGraph AI platform, cloud infrastructure (Supabase/PostgreSQL), research & development, and core product development associated costs	Fixed Cost

LLM API token consumption per AI-agent decision, IoT telemetry data bandwidth (MQTT brokers), and SMS/Email gateway notifications	Order-based / Variable Cost
The onboarding and manual integration costs for university legacy systems, B2B sales/marketing, administrative, and technical customer support services for facility managers	Overhead Cost

6.4. Key Partners

Internal	External
University IT Departments	Cloud & AI Infrastructure Providers (OpenAI, AWS)
University Student Councils (for user feedback and app adoption)	Local IoT Hardware Vendors (e.g., Cytron Technologies)

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
Multi-Agent Hallucination	High	Medium	Implement Guardrail safety checks via GitHub Actions and LangGraph state validation to intercept nonsensical AI commands.
MQTT Broker Downtime	High	Low	Deploy redundant MQTT brokers with automatic failover and use data replay features to ensure no IoT telemetry is lost during reconnection.
Unauthorized Device Control	Medium	Medium	Implement local caching for frequent requests and a fallback rule-based logic if the AI Orchestrator fails to respond.
Incorrect ESG Reporting	Medium	Low	Use automated PDF policy ingestion validation and cross-reference IoT occupancy data with physical sensor logs to ensure data integrity.

Inaccurate Booking Sync	Medium	Medium	Utilize PostgreSQL's robust state management to handle race conditions when two users attempt to book the same resource simultaneously.
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7. Implementation & Financials

7.1 Go-to-Market Strategy

SCADA-i adopts a phased, "Land and Expand" rollout strategy explicitly targeting the higher education sector. We will initiate our market entry by soft-launching at 3-5 premier universities in Malaysia (e.g., UTM, UM) acting as our "Living Labs." By providing immediate TNB energy savings and system stability during these trial runs, we will leverage the resulting case studies to aggressively scale and capture 60% of the Malaysian college and university market.

For international penetration, we will adopt a beachhead strategy: expanding first to 3 neighboring countries by securing 1-3 flagship universities per country. These early regional adopters will build our localized credibility, allowing us to rapidly expand our reach to the major Southeast Asian (SEA) market before ultimately executing a full global expansion.

7.2 Implementation Plan

Phase	Action	Timeline	Target
Pilot Launch	Deploy in 3-5 key colleges/universities in Malaysia.	Month 1-8	Validate 15% energy savings; active student/staff adoption.
National Expansion	Scale aggressively across MOHE networks (IPTA & IPTS).	Months 8-24	Capture 60% of the Malaysian higher education market.
Regional Entry	Enter 3 neighboring countries with 1-3 universities/colleges selected per country	Month 24-34	Establish footprint in key markets (e.g., Singapore, Indonesia, Thailand).
SEA Dominance & Global	Expand reach 50% of SEA market; prepare for global rollout	Month 34-60	Reach 50% SEA market share; secure first non-SEA global campus.
TOTAL	RM 3.5M+ Seed to Series A Investment	60 Months	Significant Market Share (62%)

7.3 Revenue Projections (e.g 3-Year)

Metric	Year 1	Year 2	Year 3
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<i>Active University</i>	5	40	150
<i>Connected Spatial Zones</i>	500	6300	30000
<i>Daily AI Automated Actions</i>	2000	25200	120000
<i>SaaS & Hosting Revenue</i>	RM 800,000	RM 8.5M	RM 38.2M
<i>ESaaS & Integration Revenue</i>	RM 450,000	RM 4.1M	RM 14.3M
<i>Total Revenue</i>	RM 1.25M	RM 12.6M	RM 52.5M
<i>Net Position</i>	RM (1.8M)	RM (500,000)	RM +12.5M

7.4 Key Financial Milestones

- Month 8: Successfully deployed in 3-5 key Malaysian universities; validated 15% energy savings to trigger ESaaS shared-revenue clauses and secure core student/staff adoption.
- Month 24: Captured 60% of the Malaysian higher education market (IPTA & IPTS); reaching the Break-Even point with positive contribution margins driven by domestic scaling.
- Month 34: Successful entry into 3 neighboring countries (e.g., Singapore, Indonesia, Thailand); establishing a strong regional footprint and becoming fully cashflow positive.
- Month 60: Reached 50% SEA academic market share; secured the first non-SEA global campus deployment, validating the business model for a major global expansion and/or Series B funding.